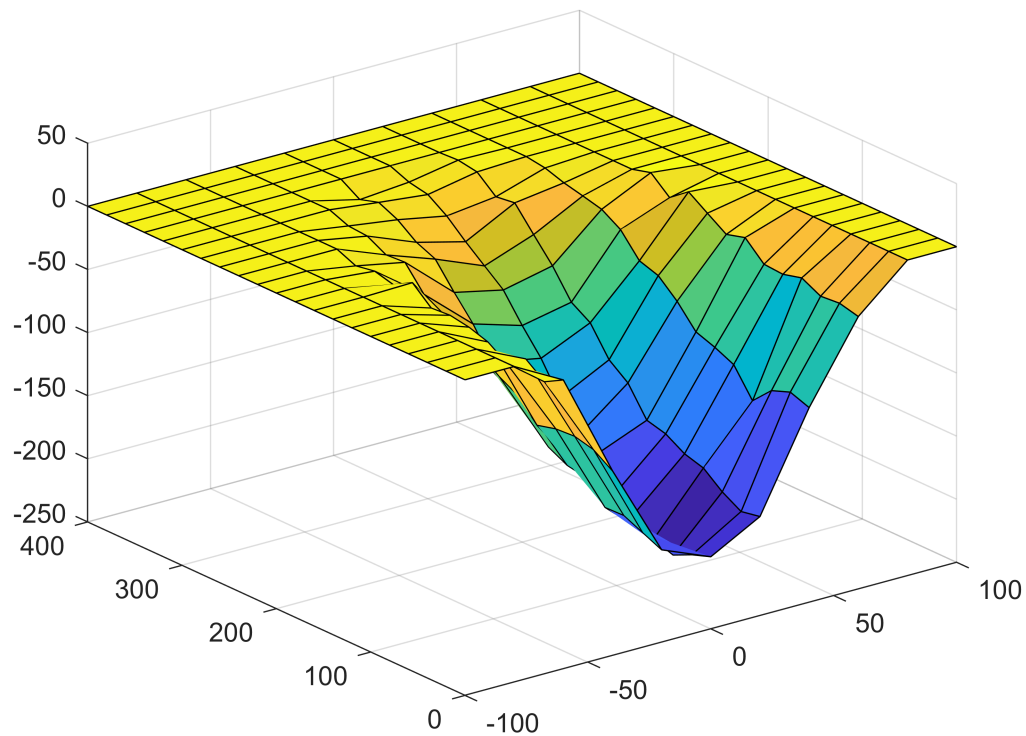


Exercise 1:

(a) Surface plot:

```
load('reservoir.mat');  
surf(X,Y,Z)
```



Part(b): computing the volume:

```
a = 200;  
b = 400;  
func = @Q;  
Volume = -integral2(func,-100,100,0,b,"Method","iterated")
```

Volume = 2.7093e+06

```
function [q] = Q(x,y)  
load('reservoir.mat');
```

```
[x1,y1] = meshgrid(X,Y);  
NewZ=interp2(x1,y1,Z,x,y);  
q=NewZ;  
end  
  
function [p] = pr(x,y)  
rho = 1000;  
g=9.8;  
p = rho*g*Q(x,y);  
end
```